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Benthic Invertebrates as Indicators of Organic Pollution in the Western Baltic Sea

Abstract

The quantitative distribution of some important benthic invertebrates in a shallow inshore area of Kiel Bay (Western Baltic Sea) is described. This region is partly polluted by domestic sewage from the municipality of Kiel. Three groups of species are distinguished with reference to their densities and other population parameters in the different subareas: Progressive species of the 1st and 2nd order indicate a high and moderate to slight degree of pollution respectively by high population numbers. Regressive species are adverse indicators, absent or occurring in exceptionally low numbers in affected areas. In many cases the examination must be concentrated on sand bottom, because a series of species normally dwelling on aufwuchs build up dense populations in sand, if it is organically enriched.

The polychaetes *Capitella capitata*, *Nereis diversicolor* and *Polydora ligni* are regarded as progressive species of the 1st order. The amphipods *Corophium insidiosum* and *Gammarus salinus*, the mussel *Mytilus edulis*, the polychaete *Pygospio elegans*, the snail *Hydrobia ulvae*, and some other invertebrates belong to the group of indicators for slight organic pollution. The main regressive species are the amphipod *Bathyporeia sarsi* and the mite *Copidognathus fabriciusi*.

Contents

1. Introduction	245
2. Methods	248
3. Results	248
3.1. Progressive species of the 1st order	249
3.2. Progressive species of the 2nd order	250
3.3. Regressive species	251
4. Conclusions and Discussion	252
5. Summary	253
6. References	253

1. Introduction

The indicator concept is based upon the assumption that relatively longlived, stationary organisms reflect the environmental conditions over a long time scale. In particular, a series of aquatic animals is regarded as closely associated with organic pollution and thus as significantly indicating this kind of man-made environmental change. A recent review of the multitude of already existing literature on this subject is given in the comprehensive report of LEPPÄKOSKI (1975).

In the present study, selected benthic invertebrates of the Western Baltic Sea were investigated for their possible value as indicator species. Their horizontal distribution in a gradient of domestic sewage pollution is described quantitatively in terms of abundance (= numerical density), biomass (dry weight), presence (= percentage of frequency), and dominance (= percentage of total population density).

Table 1. Quantitative distribution of potential indicator species in the pollution gradient. Mean ($\bar{x}$), standard error ($\pm$) and maximum (max) values of abundance (A), biomass (B), presence frequency (P) and dominance (D) given for different ranges of distance from the outlet. * = only sand data, ** = only juvenile specimens (1–10 mm body length) considered.

Species		50–200 m				300–700 m				1000–2000 m			
		A (I/m ²)	B (mg/m ²)	P (‰)	D (‰)	A (I/m ²)	B (mg/m ²)	P (‰)	D (‰)	A (I/m ²)	B (mg/m ²)	P (‰)	D (‰)
<i>Hydrobia ulvae</i>	$\bar{x}$	1212	932	92.3	4.1	1694	1468	94.0	7.7	269	118	85.0	4.7
PENNANT	$\pm$	319	236	—	1.1	436	445	—	1.5	74	39	—	0.8
	max	8590	6088	—	37.6	13710	12486	—	37.4	2725	1230	—	28.6
<i>Mytilus edulis</i> L.	$\bar{x}$	287	417	97.6	2.0	204	485	77.8	0.8	28	115	72.4	1.8
(*) (**)	$\pm$	115	133	—	1.2	99	376	—	0.2	9	52	—	0.6
	max	3400	2747	—	47.6	2607	10208	—	4.3	150	931	—	12.7
<i>Mysella</i>	$\bar{x}$	0	—	0.0	—	1.5	—	7.4	—	27	—	10.0	—
<i>bidentata</i>	$\pm$	0	—	—	—	0.7	—	—	—	18	—	—	—
MONTAGU	max	0	—	—	—	30	—	—	—	797	—	—	—
<i>Pholoe minuta</i>	$\bar{x}$	0.6	0.1	3.9	0.0	1.0	2	30.0	0.1	14	3	29.0	0.2
FABRICIUS	$\pm$	0.5	0.1	—	0.0	3	1	—	0.1	6	1	—	0.1
	max	25	5	—	0.1	100	27	—	0.8	215	82	—	1.8
<i>Anaitides</i>	$\bar{x}$	19	38	36.5	0.1	39	71	70.0	0.2	16	7	54.6	1.3
<i>mucosa</i>	$\pm$	7	16	—	0.1	10	27	—	0.0	5	2	—	1.0
BERGSTRÖM	max	260	647	—	1.8	370	880	—	1.4	230	34	—	5.5
<i>Nereis diver-</i>	$\bar{x}$	838	2828	88.5	3.1	228	918	96.0	1.6	36	149	61.8	1.6
<i>sicolor</i>	$\pm$	246	733	—	0.9	51	205	—	0.4	8	36	—	0.4
O. F. MÜLLER	max	10250	20323	—	28.6	1873	3455	—	16.1	275	1159	—	13.7
<i>Nereis succinea</i>	$\bar{x}$	4	—	7.1	0.0	11	—	25.9	0.3	0.5	—	10.0	0.1
(*)	$\pm$	3	—	—	0.0	5	—	—	0.3	0.3	—	—	0.1
LEUCKART	max	120	—	—	0.8	115	—	—	8.2	5	—	—	1.5
<i>Nephtys caeca</i>	$\bar{x}$	0	—	0.0	—	0.1	—	4.0	—	6	—	38.2	—
FABRICIUS	$\pm$	0	—	—	—	0.1	—	—	—	2	—	—	—
	max	0	—	—	—	3	—	—	—	85	—	—	—
<i>Polydora ligni</i>	$\bar{x}$	52	28	35.7	0.1	47	25	37.0	0.1	4	2	13.3	0.1
WEBSTER (*)	$\pm$	37	26	—	0.0	25	18	—	0.0	2	1	—	0.1
	max	1570	1043	—	0.8	670	479	—	0.5	35	10	—	2.6
<i>Polydora</i>	$\bar{x}$	12	5	23.8	0.0	19	5	25.9	0.0	3	0.2	16.7	0.0
<i>ciliata</i> (*)	$\pm$	6	2	—	0.0	8	2	—	0.0	2	0.1	—	0.0
JOHNSTON	max	190	59	—	0.1	205	44	—	0.2	55	3	—	0.3

<i>Pygospio elegans</i>	$\bar{x}$	17790	1550	69.2	16.1	18710	1604	100	35.3	1537	190	90.9	16.6
	$\pm$	6372	535	—	4.1	5028	498	—	3.8	487	63	—	2.7
CLAPAREDE	max	201500	18468	—	88.7	147500	15045	—	90.9	13550	2456	—	72.9
<i>Capitella capitata</i> (*)	$\bar{x}$	24744	7281	100	61.4	3416	637	96.3	26.6	187	33	83.3	13.0
	$\pm$	5818	1806	—	5.9	1265	232	—	5.8	53	9	—	2.9
FABRICIUS	max	171000	59946	—	99.6	29195	4817	—	97.1	1350	215	—	53.0
<i>Protodrilus chaetifer</i>	$\bar{x}$	0	—	0.0	—	3	—	3.3	—	67	—	5.8	—
	$\pm$	0	—	0.0	—	2	—	—	—	45	—	—	—
REMANE	max	0	—	0.0	—	100	—	—	—	2200	—	—	—
<i>Copidognathus fabriciusi</i>	$\bar{x}$	3	—	2.5	—	132	—	17.5	—	268	—	35.0	—
	$\pm$	2	—	—	—	84	—	—	—	96	—	—	—
LOHMANN	max	100	—	—	—	3300	—	—	—	3200	—	—	—
<i>Gammarus salinus</i> (*)	$\bar{x}$	29	24	69.1	1.6	126	32	44.4	0.2	31	46	30.0	5.9
	$\pm$	8	8	—	0.7	80	16	—	0.1	15	26	—	1.7
SPOONER	max	295	100	—	11.0	1923	389	—	2.6	345	696	—	9.5
<i>Bathyporeia sarsi</i> (*)	$\bar{x}$	8	3	14.3	0.2	39	18	59.3	1.0	130	67	96.7	14.4
	$\pm$	4	2	—	0.1	20	8	—	0.4	36	18	—	3.5
WATKIN	max	135	55	—	4.1	540	226	—	8.0	820	433	—	65.3
<i>Microdeutopus gryllotalpa</i> (*)	$\bar{x}$	34	6	35.7	0.0	62	13	40.7	0.1	8	1	13.3	0.1
	$\pm$	19	4	—	0.0	37	7	—	0.0	7	1	—	0.0
DA COSTA	max	580	124	—	0.3	953	190.5	—	0.8	195	34	—	1.3
<i>Corophium insidiosum</i>	$\bar{x}$	450	45	52.3	0.4	495	63	59.3	0.7	19	2	26.7	0.6
	$\pm$	359	35	—	0.2	217	35	—	0.2	11	1	—	0.4
CRAWFORD (*)	max	15070	1482	—	6.1	4205	832	—	3.3	275	45	—	11.8
<i>Idotea baltica</i> (*)	$\bar{x}$	7	5	40.5	0.0	6	6	40.7	0.0	2	3	13.3	0.1
	$\pm$	3	2	—	0.0	2	3	—	0.0	1	2	—	0.0
PALLAS	max	140	47	—	0.4	50	79	—	0.3	20	43	—	1.9

The investigation area is a shallow inshore region in the South of Kiel Bay, where high quantities of domestic sewage are discharged. Following a transect from the sewage outlet along the coast line, three zones with a decreasing degree of pollution and with characteristic faunal assemblages were found on the sandy bottom at a depth of 3 m. More detailed descriptions and references to literature on the study area are given in previous papers (ANGER, 1975a, b). The dominant group within the macrofauna with respect to both number of individuals and of species was formed by polychaetes, followed by molluscs and amphipods. The only species which are treated in this report are those which show rather distinct reactions to the environmental gradient and fulfill the main requirements for indicators recently summarized by REISH (1972, 1973) and LEPPÄKOSKI (1975).

The investigation area was divided into three subregions (50–200, 300–700 and 1000–2000 m distance from the outfall). Within each subregion three stations were sampled and the data from these three stations were then combined to give one value for each subregion.

These subareas are nearly identical with the range of the associations defined by ANGER (1975b). They correspond to LEPPÄKOSKI's "polluted", "semipolluted" and "healthy" bottoms. A "very polluted" zone *sensu* LEPPÄKOSKI is restricted to an area of less than 50 m radius around the outlet; a "semihealthy" bottom has not been delimited in this survey. The original data for all stations are given in an unpublished thesis (University Kiel, 1975).

Some of the most conspicuous trends are shown as examples in graphs. Dominance values are related to the macrofauna only; biomass is not given for meiofauna and for rare species. For some species only data from sand bottom (musselbanks and *Zostera* meadows excluded) are considered, since they show here a clearer reaction to the impact of sedimenting sewage particles (cf. ANGER, 1975a). Species which are affected only within the polluted zone (50–200 m) were regarded as broadly indifferent to organic pollution and were not included in the present study. Systematic groups not identified to the species level (e.g. Nemertini, Oligochaeta) were also excluded from considerations as well as species which are too rare. The data given in table 1 are based on a total of 157 macrofauna samples (0.2 m^2) and an equal number of observations on meiofauna.

2. Methods

Nine stations were chosen in the investigation area. Two parallel samples per station per month were taken from 1971 to 1972 by means of a 0.1 m^2 van Veen grab, strained through a sieve with 1 mm mesh size and fixed in 4% formalin solution. The material was sorted under a stereo microscope, biomass was calculated as dry weight (100 °C to weight constancy, including hard parts). Additionally, one unstrained parallel sample was taken at each station with a smaller grab (0.01 m^2) in order to gain information on meiofauna and young bottom stages. For further information on aims and methods and on some chemical characteristics of the sewage see ANGER (1975b).

3. Results

The most significant population parameters determined in this study are given in Table 1. Following the definitions by LEPPÄKOSKI (1975) we may distinguish here three groups of species with reference to their distribution pattern in the study area. This pattern is regarded as an expression of their response to mainly organic pollution of the benthic environment.

3.1. Progressive species of the 1st order

The polychaete *Capitella capitata* is a typical indicator. In the full range of its worldwide distribution it is known to react most clearly to organic pollution (for literature see AUGUSTIN & ANGER, 1974, LEPPÄKOSKI, 1975). It develops the highest population density and biomass in the polluted zone (fig. 1) and there it dominates, often

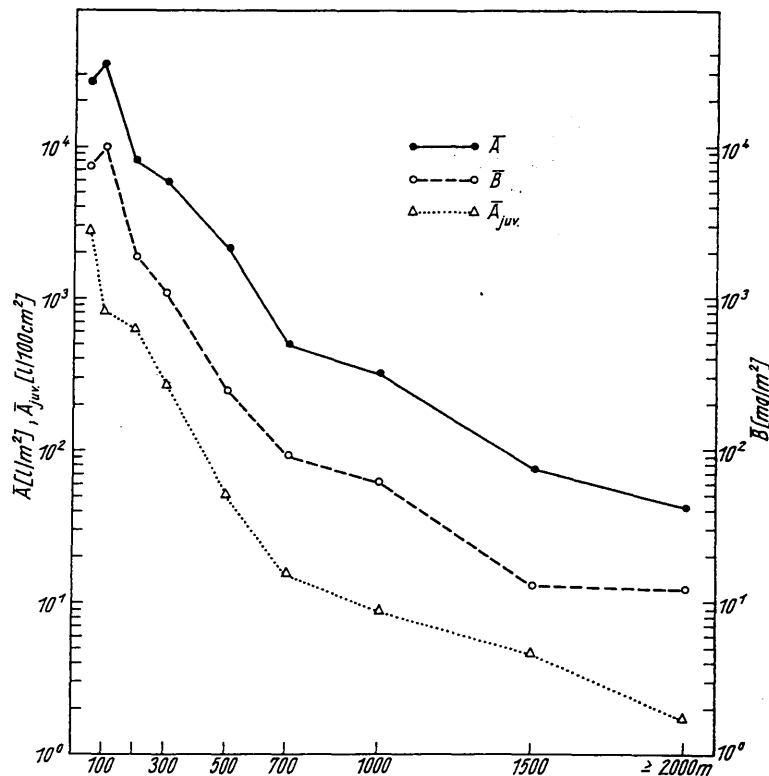


Fig. 1. *Capitella capitata*: Mean abundance ($\bar{A}$), biomass ($\bar{B}$) and abundance of juvenile bottom stages ($\bar{A}_{juv}$) in the pollution gradient (distance from the outlet).

In Fig. 1, ordinate, please read $\bar{A}$ [1/m²], $\bar{A}_{juv}$ [1/100 cm²] (instead $\bar{A}$ [1/m²], $\bar{A}_{juv}$ [1/100 cm²]).

with a percentage of more than 99% of the total macrofauna. Only bacteria, ciliates and nematodes (and at the outer station (200 m) often the polychaete *Pygospio elegans*) are more numerous in the polluted part of the study area. Corresponding to the literature, there seems no doubt that *Capitella capitata* is an appropriate indicator for organically enriched marine bottoms.

The population parameters of the polychaete *Nereis diversicolor* are also multiplied in the polluted area (fig. 2). As it is normally rather scarce in clean sand, it must be regarded as belonging to this group of indicators. The present results correspond well with those of many other authors in the Baltic Sea region (for review see LEPPÄKOSKI, 1975).

Like some other *Polydora* species, *P. ligni* is closely associated with a high amount of suspended organic matter. In particular in the region 200–300 m from the sewage pipeline its density and biomass are increased by a factor of 100. It can also occur in

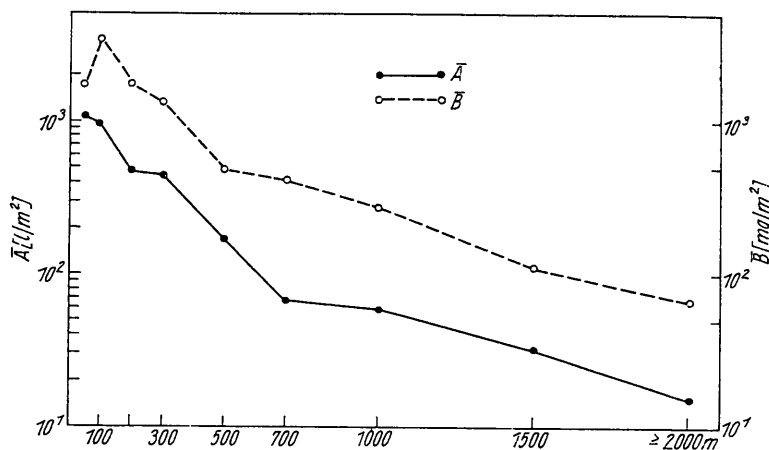


Fig. 2. *Nereis diversicolor*: for explanation see fig. 1.
Please read $\bar{A}$ [I/m^2] (ordinate) instead $\bar{A}$ [l/m^2].

mass populations on aufwuchs and peat without sewage pollution. In pure sandy bottoms, which are normally avoided by the polychaete, it can only maintain high numbers if a fairly high precipitation of digestible substances takes place. This observation, already recorded by ANGER (1975a), holds true also for some other euryoeceous aufwuchs dwellers treated below.

3.2. Progressive species of the 2nd order

Some macrobenthic invertebrates are so strongly favoured by sewage pollution that they could also belong to the first group. Mainly the amphipod *Corophium insidiosum*, the mussel *Mytilus edulis*, and the polychaete *Pygospio elegans*, but to a certain extent

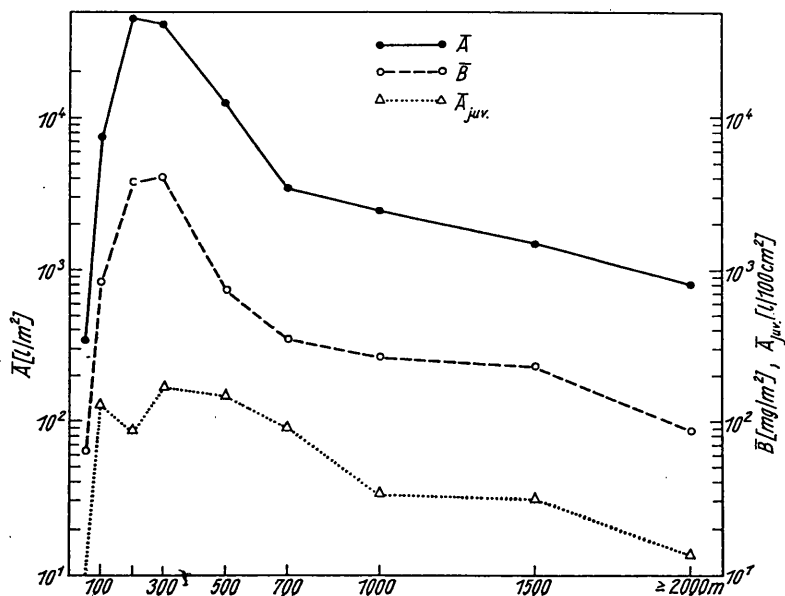


Fig. 3. *Pygospio elegans*: for explanation see fig. 1.
Please read $\bar{A}$ [I/m^2] and $\bar{A}_{juv}$ [$I/100 m^2$] (ordinates).

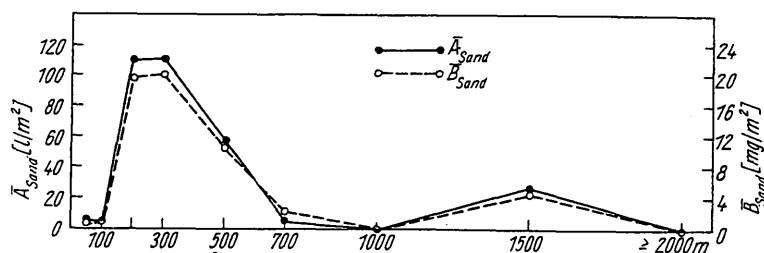


Fig. 4. *Microdeutopus gryllotalpa*: for explanation see fig. 1. Only sand data considered. Please read $\bar{A}_{\text{Sand}} [1/m^2]$ (ordinate) instead $\bar{A}_{\text{Sand}} [1/m^2]$.

also, the polychaetes *Polydora ciliata* and *Nereis succinea* and the isopod *Idotea baltica* show a strong rise in both the semipolluted and polluted subareas. Since this rise is more pronounced in the former region and at the same time there is a clear decrease in the inner part of the latter one they are put into this second group. Typical distribution patterns are shown in fig. 3 and 4 for *Pygospio elegans* and the amphipod *Microdeutopus gryllotalpa*. More representatives of progressive species of the 2nd order are: the snail *Hydrobia ulvae*, the polychaete *Anatides mucosa*, and the amphipod *Gammarus salinus* (and less clearly, *G. oceanicus*). The bivalves *Mya arenaria* and *Macoma baltica* also have somewhat higher numbers in the slightly polluted zone. They illustrate an occurrence which is most conspicuous for *Mytilus edulis*: After the spatfall of larvae, an enormous mortality of young stages can be observed on sandy bottoms. It is highest at a size of 1 mm or less. All numbers given for *Mytilus edulis* in table 1 concern only individuals of a size 1 to 10 mm, which have survived the critical phase on vegetation-free sand bottom. Apparently, in the semipolluted zone the survival is significantly higher due to a sufficient food supply.

These facts confirm again the assumption mentioned above that many growth dwellers with potential indicator value show their most conspicuous responses to organic pollution in habitats where they are normally found only infrequently or in small numbers. LEPPÄKOSKI (1975) and other authors reviewed by him observed for *Nereis diversicolor* and *Macoma baltica* an increase of individual weight in polluted zones. The same trend has been found in the present study for several polychaetes, the snail *Hydrobia ulvae* and for *Corophium insidiosum* (for the latter species see ANGER, in press). The question is raised, whether the main reason is a higher growth rate due to a better food supply or a steady immigration of older (less sensitive) stages. The former is supposed to have a stronger effect, but evidence is yet to be provided by special experiments.

3.3. Regressive species

LEPPÄKOSKI's subgrouping into regressive species of 1st and 2nd orders could not be followed in this study, because it proved too restrictive to be applied to the material treated here. Since no species were found to be totally absent from all polluted parts of the investigation area, the following species correspond rather to regressive species of the 2nd order: *Bathyporeia sarsi* (amphipod), *Pholoë minuta*, *Nephtys caeca*, *Prodrilus chaetifer* (polychaetes), *Cepidognathus fabriciusi* (mite), *Mysella bidentata* (bivalve). Some of these species belong to the meiofauna, most of them occur rather scarcely even in undisturbed environments. Many additional infrequent species (e.g. *Bathyporeia pilosa* or the mysid *Gastrosaccus spinifer*) have a similar distribution pattern indicating a negative correlation with the degree of pollution. As „adverse indicators“ they should be absent from affected but common in unaffected regions. This second precondition is only met satisfactorily in the investigation area by the

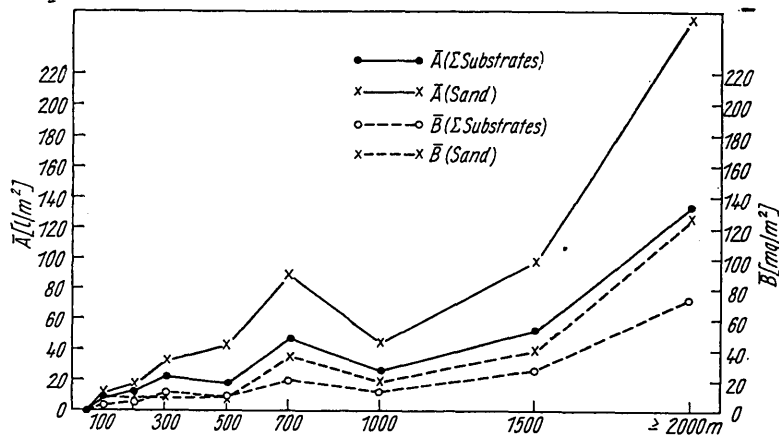


Fig. 5. *Bathyporeia sarsi*: for explanation see fig. 1. Sand data and combined data obtained from all substrates considered separately. Please read $\bar{A}$ [I/m²] (ordinate) instead $\bar{A}$ [I/m²].

sand dwellers *Bathyporeia sarsi* (fig. 5) and *Copidognathus fabriciusi*. They can be considered as suitable indicators for rather clean shallow sand bottoms in the Western Baltic Sea.

4. Conclusions and Discussion

The species treated in this report can indicate different degrees of organic pollution in the reach of the Western Baltic, in a few cases also for further areas. They can be divided into three groups according to their response to sedimenting sewage particles and the subsequent environmental changes (Tab. 2). Most of them should be studied on aufwuchs-free sand, as they have clearer reactions on this substrate if affected by organic enrichment.

These results are in fairly good accordance with those of a preliminary pilot study (ANGER, 1975a). In some cases there are differences which can be explained partly by the simple methods in combination with the purely qualitative statistics used for the pilot study. Nevertheless it could be shown that benthic organisms are able to serve as indicators of environmental quality. Many other authors (see LEPPÄKOSKI, 1975)

Table 2. Groups of benthic indicators for organic pollution. In parenthesis: species relatively rare on the examined substrates in the investigation area

Indicators for organic pollution (progressive species of 1st order)	Indicators for slight organic pollution (progr. species of 2nd order)	Adverse indicators (Regressive species)
<i>Capitella capitata</i> <i>Nereis diversicolor</i> <i>Polydora ligni</i>	<i>Corophium insidiosum</i> <i>Mytilus edulis</i> <i>Pygospio elegans</i> <i>Hydrobia ulvae</i> <i>Gammarus salinus</i> (<i>Polydora ciliata</i>) (<i>Nereis succinea</i>) (<i>Idotea baltica</i>) (<i>Microdeutopus gryllotalpa</i>) (<i>Anaitides mucosa</i>)	<i>Bathyporeia sarsi</i> <i>Copidognathus fabriciusi</i> (<i>Pholoe minuta</i>) (<i>Nephtys caeca</i>) (<i>Protodrilus chaetifer</i>) (<i>Mysella bidentata</i>)

found very similar reactions for the species treated here. However some disagreements also occur in the literature: e.g. JÄRVEKÜLG (1970) mentioned *Mytilus edulis* as oligosaprobic, while WILHELMI (1916) called it indifferent (like ANGER, 1975a, after qualitative studies), ROSENBERG (1971) found it receding after organic contamination of the Saltkällefjord, and SMYTH (1968) described it as frequent in polluted parts of the Firth of Forth.

In the present study it proved ultimately favoured by organic enrichment. For some species the larvae might be the key for understanding their distribution: *Polydora ciliata* for example is known as an inhabitant of polluted harbours, fjords and inlets (e.g. PERSOONE & PAUW, 1968, STIRN, 1972, LEPPÄKOSKI, 1975) as are the larvae (SMIDT, 1944, SCHRAM, 1968). The settlement of their planktonic stages is favoured by bacterial films on the sediment (DORSETT, 1961, KISELEVA, 1967), which can also be related to organic pollution. These examples emphasize the need for more detailed experimental and field studies on the biology and ecology of both the single species and the communities for a better understanding of the ecosystem and its responses to human and natural perturbations. The indicator concept is one of many helpful techniques to characterize an environment and to make predictions of its future behaviour under rising stress.

5. Summary

Three groups of indicator species are distinguished on the basis of their distribution pattern in a pollution gradient observed in the south of Kiel Bay (Western Baltic Sea). Three polychaetous annelids show distinct indicator qualities by inhabiting polluted sand bottoms in high numbers. Some other benthic invertebrates occur mainly under conditions of slight organic pollution; a few species can be regarded as very sensitive, thus indicating pollution by their absence or exceptionally low numbers.

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